

Systems and Internet Security Technologies

Lecture 2
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Lecture 2

Insider Threat and Ransomware

Insider threat

Most elusive threats to an organization comes from within!

Causes:

- Malicious intent (revenge)
- By mistake
- Negligence



INSIDER THREAT

Insider threat manifestation:

- Malicious insiders: who intentionally harm the company.
- Negligent insiders: who inadvertently cause damage through carelessness.
- Accidental insiders: who fall victims to phishing or social engineering attacks.

Preventing Insider threats

How to prevent insider threats?

- Technical controls such as access management and user behavior analytics.
- Organizational measures like regular training and a culture of security awareness:
 - Encouraging employees to report suspicious activities.
 - Ensuring they understand the protocols for handling sensitive information.

Ransomware

What is Ransomware?

- Malicious software that encrypts files on a device or network, holding them hostage until a ransom is paid. Usually in some form of cryptocurrency.



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Double extortion

Ransomware attackers often threaten to release sensitive stolen data or report you to regulators if the ransom is not paid, increasing the pressure to comply.



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How Ransomware Spreads?

- Ransomware infiltrates into systems in the same way as other malicious codes do.
- For example, using phishing emails, exploiting vulnerabilities, or malicious downloads.

How to protect against ransomware?

• **Foundational Cybersecurity Measures**

- Regular software updates. (To patch vulnerabilities)
- Robust endpoint security solutions. (Using robust antivirus and anti-malware tools)
- Educating employees about phishing. (Training employees to recognize phishing attempts and other threats)

How to protect against ransomware?

• **Specific Ransomware Countermeasures**

- Immutable storage architectures.
- Application whitelisting:
- Restricting administrative
- Network segmentation

Advanced defense tactics

- End-to-end encryption
- Incident response
- Threat hunting activities
- Threat intelligence services



This is an application of the Defense in Depth strategy. Combining general malware defenses with ransomware-specific tactics offers the best protection. It's not just about resisting attacks but also recovering quickly when they occur.